

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT Plan the learning first; let the checklist follow. Click a field to type • click a box to tick it.

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbially	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 10 (25-29 Oct 2026) - Period 1	Unit 3 - 3.1 Web Interface for the Game (1/2)
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 10 / Lesson 1 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Unit 2 test avg 76%; selection tools weakest. Short re-teach sheet sent. New unit: start from students' game ideas from homework.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students set up the 'Bit by Bit' worm-game project: folder structure, naming (camel/kebab case), VS Code, and the HTML structure of the index page. Sprint 3.1 obj. 1; ISTE 4.a; Mass. 9-12.DTC.a.1; CSTA 3A-AP-16.

Success criteria — what will successful learning look / sound like?

I create the correct folder structure (html, css, js, images, audio); I name files in kebab case; I write a valid index.html with head, body, header, sections and form elements.

Key knowledge / skill / understanding:

HTML structures content; folders keep projects organised; naming conventions avoid broken links.

Likely misconception / access or challenge need:

Spaces in file names are fine; HTML styles the page; EAL: 'hierarchy', 'kebab case'.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Brainstorming ☒ Time: 7 min	Strategy: Modelling / worked exam ☒ Time: 22 min	Strategy: Think-pair-share ☒ Time: 8 min
What students will do: Groups brainstorm what a 'catch the worm' vermicompost game needs on the start page (title, player name, glove choice, start button) and sketch the layout.	What students will do: Download the resource folder; create folders and kebab-case files; in VS Code write index.html structure (doctype, head, title, header, main, form, input, button) following the teacher.	What students will do: Pairs examine 3 file names and 3 code snippets; decide which follow camel/kebab case and which HTML has errors, then share fixes.
Evidence tool: Observation checklist ☒ Assessment: ☒ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☐ Pair ☒ Group	Evidence tool: Practical / performance task ☒ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Mini-whiteboards ☒ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group
Resources: A3 paper/markers; vermicompost picture (real-life link).	Resources: VS Code; resource folder go.qubitcs.com/cXAd; QR LCdB; source file h9pt.	Resources: Slide with snippets; mini-whiteboards.
Support (low performers / SEN): Picture prompts of game screens.	Support (low performers / SEN): Starter index.html with comments marking where to type.	Support (low performers / SEN): Highlighted lines to focus on.
Stretch (fast learners / gifted): Add an accessibility feature to the sketch.	Stretch (fast learners / gifted): Add semantic tags and alt text; validate the page with the W3C validator.	Stretch (fast learners / gifted): Rewrite a snippet using semantic HTML.

3 PREPARE — What do I need to do so students can succeed?				4 RESPOND — What will I do with what I see and hear?			
Set students up to succeed: ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources ☒ Praise, recognition and reinforcement planned				If students struggle or finish early I will: ☒ Re-teach ☒ Regroup ☒ Give extra support ☒ Stretch ☒ Have students redo / improve			
My setup / modelling: Time: 6 min Hook: play the finished worm game for 1 minute. Real-life link: vermicompost (earthworms recycle food waste - school garden). Model folder setup and VS Code basics (Explorer, Live Server/open in browser). Key words: camel case, kebab case, hierarchy, element, tag.				What I will watch for / likely responses: Files saved in Downloads not the project folder; missing closing tags; uppercase/space file names; students copying the source file without understanding. Response: 'my page is blank' - file not saved/opened.			
What is new or innovative in this lesson? <i>(ETEC looks for this)</i> Students build a real browser game across Units 3-4, linked to sustainability (vermicompost) - a project that grows each lesson.				Close / exit check <i>(summative + time for learners' questions)</i> Time: 7 min Exit ticket: write the correct folder tree and one kebab-case file name; one question for the teacher. Learners' questions.			
Progress recorded / shared: ☒ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio				Homework / platform <i>(e.g. Madrasati)</i> Madrasati: complete Challenge 1 page-structure sketch for a personal website (drawing + 5 HTML tags you would use).			

5 REFLECT — After the lesson <i>Complete within a day. This is reflective-practice evidence and feeds your next Starting point.</i>							
What worked — and how do I know? Game demo hooked everyone; vermicompost link sparked talk about recycling practices in different home countries. Folder structure done by 90%.			What will I change next time? Check VS Code is installed on all lab PCs before the unit (lost 5 minutes in 10/4). Keep the starter file for SEN.			Students needing follow-up: Students new to coding (mainly recent arrivals) need a 'tags cheat sheet'; 3 students did not save in the right folder.	

COVERAGE CHECK <i>Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.</i>							
Literacy read + write · 2-1-1-6 ☐1 ☒2 ☐3 ☐C ☐H	Numeracy numbers / data · 2-1-1-6 ☐1 ☐2 ☐3 ☐C ☒H	Digital students use it · 2-1-1-4 ☐1 ☒2 ☐3 ☐C ☒H	Digital research trusted sources · 2-1-1-4 ☐1 ☒2 ☐3 ☐C ☐H	Real-life ① application · 2-1-1-5 ☒1 ☐2 ☐3 ☐C ☐H	Real-life ② application · 2-1-1-5 ☐1 ☒2 ☐3 ☐C ☐H	Inquiry experiment / research · 2-1-1-7 ☒1 ☐2 ☐3 ☐C ☐H	Thinking problem-solving · 2-1-1-7 ☐1 ☐2 ☒3 ☐C ☐H
Create something new · 2-1-1-7 ☒1 ☒2 ☐3 ☐C ☐H	Discussion expressing ideas · 2-1-1-8 ☒1 ☐2 ☐3 ☐C ☐H	Pair / group cooperative work · 2-1-1-8 ☒1 ☐2 ☒3 ☐C ☐H	Learner choice independent work · 3-2-1-6 ☐1 ☐2 ☐3 ☐C ☒H	Self-assessment 2-2-1-1 ☐1 ☐2 ☐3 ☒C ☐H	Peer assessment 2-2-1-1 ☐1 ☐2 ☒3 ☐C ☐H	SEN / low specific support · 2-1-1-3 ☒1 ☒2 ☒3 ☐C ☐H	Gifted specific stretch · 2-1-1-3 ☒1 ☒2 ☒3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY:	Safe, respectful climate	•	every learner drawn in	•	purposeful use of time	•	smooth transitions	•	learner voice	•	specific feedback and next steps	•	students act on feedback
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Mr. Ashraf Elbially	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 10 (25-29 Oct 2026) - Period 2	Unit 3 - 3.1 Web Interface for the Game (2/2)
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 10 / Lesson 2 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Exit ticket: 88% correct folder tree; closing-tag errors common. Today: start with a 'spot the bug' and build the game page.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students build the HTML structure of the game page (score, worm count, holes grid, tray, gloves, buttons) and plan a page structure from a layout. Sprint 3.1 obj. 1; Challenges 2-3; CSTA 3A-AP-17.

Success criteria — what will successful learning look / sound like?

I break a layout into header/container divs; I write the game page HTML with ids/classes for each part; my page opens without errors.

Key knowledge / skill / understanding:

div containers, id vs class, nesting creates hierarchy; plan structure before coding.

Likely misconception / access or challenge need:

id and class are the same; every element needs its own CSS file.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Problem solving ☒ Time: 8 min	Strategy: Modelling / worked exam ☒ Time: 22 min	Strategy: Jigsaw ☒ Time: 7 min
What students will do: Challenge 2: convert a given web page screenshot into a page-structure diagram (header, innerHeader1/2, container, innerContainer1/2).	What students will do: Code game.html: header with score and worm count, a holes container with 9 holes (div), worm tray, glove area, and buttons, each with ids/classes as in the source file.	What students will do: Expert groups each explain one part of the page (header, holes, tray, gloves) to a mixed group, using their code.
Evidence tool: Open-ended task ☒ Assessment: ☒ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group	Evidence tool: Practical / performance task ☒ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Oral questioning ☒ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☐ Pair ☒ Group
Resources: Printed screenshot; diagram template.	Resources: VS Code; QR nCfN; source file F9aL.	Resources: Each student's code; jigsaw cards.
Support (low performers / SEN): Template with boxes pre-drawn.	Support (low performers / SEN): Partially completed game.html; colour-coded nesting guide.	Support (low performers / SEN): Expert role card with prompts.
Stretch (fast learners / gifted): Add class names using camel case.	Stretch (fast learners / gifted): Use a loop-friendly naming pattern (hole1-hole9) and comment each section.	Stretch (fast learners / gifted): Explain how this structure will help JavaScript later.

3 PREPARE — What do I need to do so students can succeed?	4 RESPOND — What will I do with what I see and hear?
Set students up to succeed: ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources ☒ Praise, recognition and reinforcement planned	If students struggle or finish early I will: ☒ Re-teach ☒ Regroup ☒ Give extra support ☒ Stretch ☒ Have students redo / improve
My setup / modelling: Time: 6 min Spot the bug: 3 HTML lines with missing closing tags. Model planning on paper first: draw boxes for the game page, then translate into nested divs. Explain id (unique) vs class (shared). Share success criteria.	What I will watch for / likely responses: Mis-nested divs; duplicate ids; students not linking files later; copying without understanding - ask 'what will this div hold?'. Likely response: 'Can I just use class for everything?'
What is new or innovative in this lesson? <i>(ETEC looks for this)</i> Jigsaw: every student becomes an 'expert' explaining one part of the game structure - builds learner voice and collaboration across language groups.	Close / exit check <i>(summative + time for learners' questions)</i> Time: 7 min Exit check: label the structure diagram of the game page with correct tags; learners' questions.
Progress recorded / shared: ☒ Gradebook ☐ Parents ☒ Written feedback in notebook / portfolio	Homework / platform <i>(e.g. Madrasati)</i> Madrasati: Challenge 3 - draw a responsive image-gallery layout and write its HTML (no CSS yet).

5 REFLECT — After the lesson <i>Complete within a day. This is reflective-practice evidence and feeds your next Starting point.</i>		
What worked — and how do I know? Planning on paper first reduced nesting errors a lot. Jigsaw groups mixing Arabic, Urdu and English speakers worked; experts were proud to explain.	What will I change next time? Give more time for the game page; 30% needed the starter file. Shorten jigsaw to 2 rounds.	Students needing follow-up: 10/2 and 10/6: 4 students with duplicate ids to fix. Gifted coders could start the gallery challenge in class.

COVERAGE CHECK <i>Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.</i>							
Literacy read + write · 2-1-1-6 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Numeracy numbers / data · 2-1-1-6 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Digital students use it · 2-1-1-4 ☒ 1 ☒ 2 ☐ 3 ☐ C ☒ H	Digital research trusted sources · 2-1-1-4 ☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	Real-life ① application · 2-1-1-5 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Real-life ② application · 2-1-1-5 ☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	Inquiry experiment / research · 2-1-1-7 ☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	Thinking problem-solving · 2-1-1-7 ☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H
Create something new · 2-1-1-7 ☐ 1 ☒ 2 ☐ 3 ☐ C ☒ H	Discussion expressing ideas · 2-1-1-8 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Pair / group cooperative work · 2-1-1-8 ☒ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Learner choice independent work · 3-2-1-6 ☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	Self-assessment 2-2-1-1 ☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	Peer assessment 2-2-1-1 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	SEN / low specific support · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H	Gifted specific stretch · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback
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